

Decision Log

Team: HackForce (HM26-3E0C) | Sub-problem: AI-assisted category detection & description from complaint photos | Date: 20 Sept 2026

Q1. What approach did we take, and what did we reject?

Our approach: AI-suggested category and description, generated straight from the complaint photo and shown to the citizen as an editable draft before submission.

When a citizen uploads a photo, we send it to a vision model with a prompt constrained to our five categories (pothole, garbage, streetlight, illegal dumping, blocked drainage) and ask for a one-line caption. The model's top category and caption pre-fill the category dropdown and description field. The citizen can accept or edit either before filing — nothing is auto-submitted without their review.

Alternative we considered and rejected: keep the manual form we already had working — a plain category dropdown plus a free-text description box, filled entirely by the citizen. It was our obvious first build: simple, reliable, no external dependency, and what we shipped first while the rest of the app came together.

Q2. Why did we reject it? The trade-off

Dimension	Our approach (AI-suggested)	Rejected alternative (manual only)
Speed of filing	Faster — most fields pre-filled	Slower — citizen types everything
Works on patchy connectivity	Needs a network call to classify	Works fully offline
Consistency across reports	More consistent category labels	Inconsistent, citizen-dependent wording
Build effort in 72 h	Higher — needs vision API + fallback	Lower — already built, working

We accept that the AI step depends on network connectivity and adds a few seconds of latency before the form fills in — a real cost in a city with patchy signal. That is why we kept the manual form underneath as a live fallback: if the AI call fails, times out, or the citizen is offline, they fill the category and description by hand exactly as before, and filing is never blocked on the AI working.

Q3. What breaks at the scale of all of Mysuru?

What breaks first	Why (rough number)	How we'd fix it
Vision API calls at citywide volume	Thousands of photos/day, worse during Dasara spikes → rate limits, cost, latency	Queue/batch requests; cache results for visually near-duplicate reports (same pothole, many photos)
Classification over patchy/offline connections	A citizen mid-upload with no signal can't reach the AI service	Make classification asynchronous — file immediately with a manual/default category, classify in the background once connectivity returns
Night/low-light dumping photos	Most illegal dumping happens at night per the brief — accuracy drops on grainy, low-light images	Route low-confidence classifications to manual officer review instead of auto-accepting them

The single change we'd make first: make the AI step asynchronous and non-blocking, since that's what keeps reporting usable in exactly the low-connectivity conditions this city actually has.